

Complete Case Analysis

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [3]: df = pd.read_csv('data_science_job.csv')
df.head()
```

```
Out[3]:
```

	enrollee_id	city	city_development_index	gender	relevent_experience	enrolled_univ
0	8949	city_103	0.920	Male	Has relevent experience	no_enrol
1	29725	city_40	0.776	Male	No relevent experience	no_enrol
2	11561	city_21	0.624	NaN	No relevent experience	Full time c
3	33241	city_115	0.789	NaN	No relevent experience	
4	666	city_162	0.767	Male	Has relevent experience	no_enrol

```
In [6]: df.isnull().mean()*100
```

```
Out[6]: enrollee_id          0.000000
city          0.000000
city_development_index    2.500261
gender        23.530640
relevent_experience      0.000000
enrolled_university      2.014824
education_level    2.401086
major_discipline    14.683161
experience          0.339284
company_size       30.994885
company_type       32.049274
training_hours      3.998330
target            0.000000
dtype: float64
```

```
In [8]: cols = [var for var in df.columns if df[var].isnull().mean() < 0.05 and df[var].isn
cols
```

```
Out[8]: ['city_development_index',
'enrolled_university',
'education_level',
'experience',
'training_hours']
```

In [9]: `df[cols].sample(5)`

Out[9]:

	city_development_index	enrolled_university	education_level	experience	training_h
15197	0.920	no_enrollment	High School	3.0	
18373	0.624	Full time course	Graduate	3.0	
18681	0.939	Full time course	Graduate	8.0	
309	0.913	no_enrollment	Graduate	2.0	1
8734	0.899	no_enrollment	Graduate	20.0	

◀  ▶

In [10]: `len(df[cols].dropna()) / len(df)`

Out[10]: 0.8968577095730244

In [11]: `updf = df[cols].dropna()
updf.shape`

Out[11]: (17182, 5)

In [12]: `df.shape`

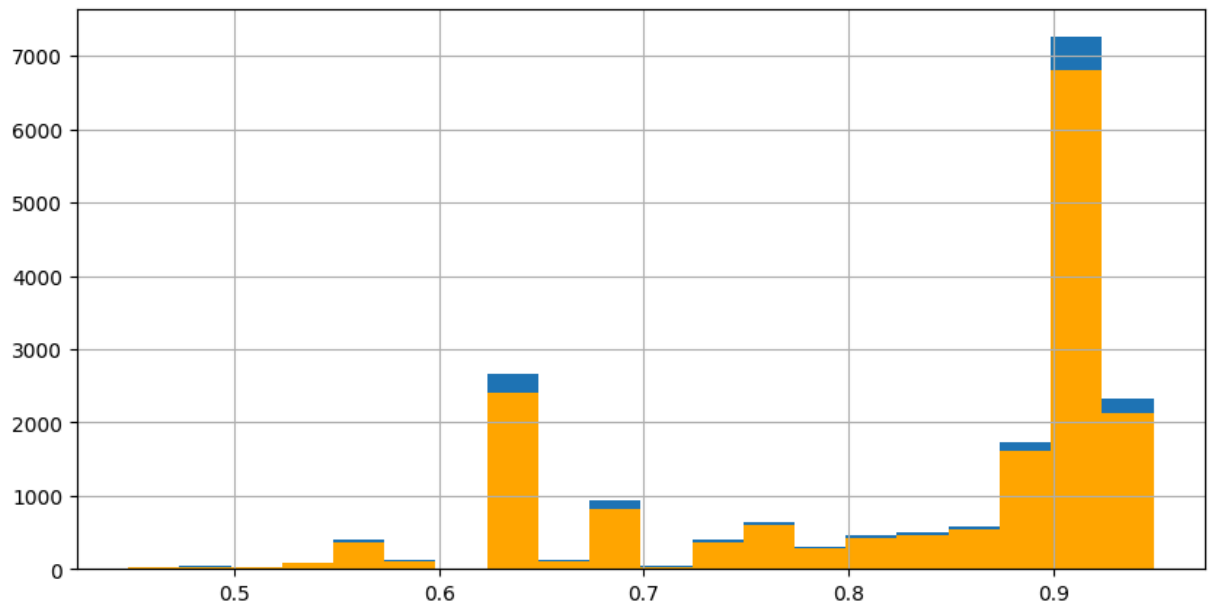
Out[12]: (19158, 13)

In [14]: `fig = plt.figure(figsize=(10, 5))
ax = fig.add_subplot(111)

df['city_development_index'].hist(bins=20, ax=ax)

updf['city_development_index'].hist(bins=20, ax=ax, color='orange')`

Out[14]: <Axes: >



```
In [15]: new = pd.concat([df['education_level'].value_counts()/len(df), updf['education_level'].value_counts()/len(updf)], axis=1)
new.columns = ['original', 'updated']
new
```

Out[15]:

	original	updated
Graduate	0.605387	0.619835
Masters	0.227633	0.234082
High School	0.105282	0.107380
Phd	0.021610	0.022116
Primary School	0.016077	0.016587

In []: